Integers

This is a collection of tasks with integers, most of the tasks are about checking if number is prime. This is an introduction to the fact that there are faster and slower methods.

For a benchmarking task you should look for time measurement library for you language. For example in python is is time, in c it is time.h and in c++ you should use chrono.

Contents

Palindrome

You are given a number **n**, check if this is palindrome.

Input Format

n - integer value.

Standard input

Output Format

boolean value.

Standard output

Examples

```
121
```

```
true
```

```
2342241324
```

```
false
```

Inside number

Calculate a count of digits and a sum of digits in a given number **n**.

Input Format

n - integer value.

Standard input

Output Format

Count of digits: integer value.

Sum of digits: integer value.

Standard output

Examples

```
111
```

```
3
3
```

```
9801
```

```
4
18
```

Is Divisible by?

Check if given **n** is divisible by numbers in range from 2 to **n** - 1.

Input Format

n - integer value.

Standard input

Output Format

A list of boolean values.

Standard output

Examples

```
5
```

```
[false, false, false]
```

```
10
```

```
[true, false, false, true, false, false, false, false]
```

Divisors count

Use your Task 3 result and count divisors of a number.

Input Format

n - integer value.

Standard input

Output Format

Count of divisors : integer value.

Standard output

Examples

```
5
```

```
0
```

```
10
```

```
2
```

Is Prime?

Use your Task 4 result to find if number is prime.

Input Format

n - integer value.
Standard input

Output Format

boolean value.
Standard output

Examples

```
5
```

```
true
```

```
10
```

```
false
```

Do I need a list?

Check if number is prime, not using a list.

Input Format

n - integer value.

Standard input

Output Format

boolean value.

Standard output

Examples

```
5
```

```
true
```

```
10
```

```
false
```

Is Prime? Going faster

Do you need to check from 2 to **n** - 1, maybe you can lower the range?

Input Format

n - integer value.
Standard input

Output Format

boolean value.
Standard output

Examples

```
5
```

```
true
```

```
10
```

```
false
```

Is Prime? Sieve of Eratosthenes

Collect all prime numbers that are less than **n**. You can google algorithm by name.

Input Format

n - integer value.
Standard input

Output Format

A list of prime numbers.
Standard output

Examples

```
5
```

```
2 3
```

```
20
```

```
2 3 5 7 11 13 17 19
```

Benchmarking

You can wrap up all your prime number search solutions in functions and measure time of their work. You sould test them on random values, multiple time for each function and calculate the mean time.

For this you should rework Sieve algorithm and make it also answering a question if given number is prime. It will be slow by the way, but you can think of how to cache your calculations.

You can leave this task for later if you feel not confident enough.